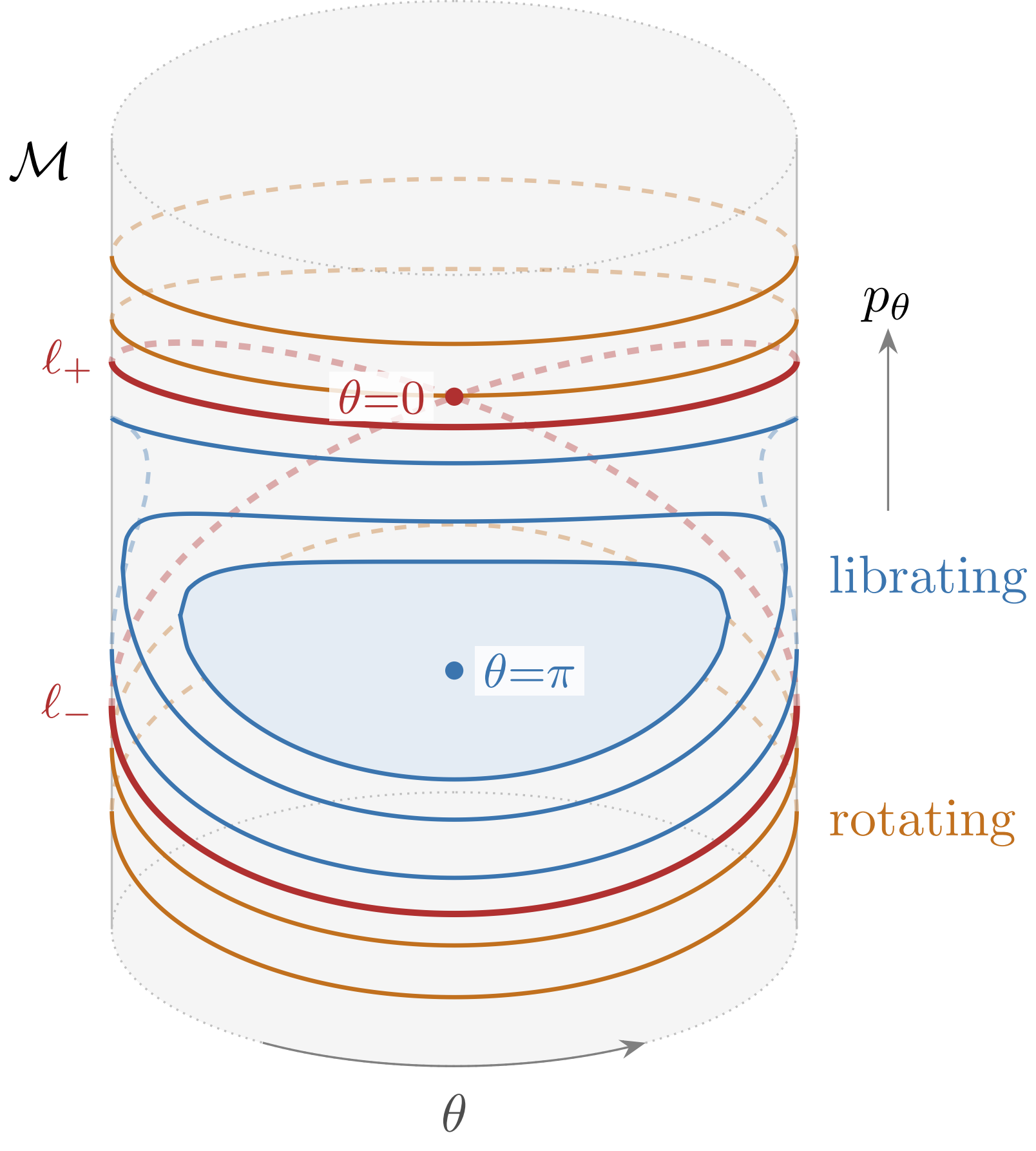

$\mathcal{M}$
p_θ
ℓ_+
$\theta=0$
librating
$\theta=\pi$
ℓ_-
rotating
θ